

Overfit-Aware Signals

Evaluating cross-sectional equity factors with DSR and PBO

Venkata Anirudh Devireddy | endokelp.com | Python research package

What it is

Most backtests report one Sharpe ratio and stop. Under multiple testing that number is easy to inflate. This package runs three real cross-sectional signals, 12-1 momentum, one-month reversal, and low-vol, through purged CV, combinatorial purged CV, Deflated Sharpe Ratio (Bailey and Lopez de Prado, 2014), and Probability of Backtest Overfitting via CSCV. The product is a per-signal PASS or FAIL verdict under fixed thresholds, not a vanity Sharpe chart.

Results (live panel demo - non-claim / survivorship-biased)

50 liquid US equities, monthly rebalance, 2005-01 to 2026-06, long-only top-5, 10 bps one-way costs. DSR threshold ≥ 0.95 , PBO threshold ≤ 0.05 . PBO is set-level. Table uses `n_trials = 3`; research log = 9. Absolute Sharpes are optimistic illustration, not a deployment claim. Prefer `synth` offline.

signal	Sharpe	CAGR	median block Sharpe	DSR	PBO	verdict
momentum	0.921	0.183	0.925	0.999	0.041	PASS
reversal	0.453	0.081	0.526	0.860	0.041	FAIL
lowvol	0.798	0.100	0.831	0.994	0.041	PASS

Methods

- Purged K-Fold with embargo (AFML ch. 7): purge on look-back-window overlap, not only on labels
- CPCV: $S = 8, k = 2 \rightarrow 28$ combinatorial block Sharpes; $\varphi = 7$ paths (collapse for fixed rules)
- DSR = PSR(SR0) using non-excess kurtosis (pandas `kurt + 3`); CSCV ranks by Sharpe
- PBO via CSCV, reject if $PBO > 0.05$

Non-claim

Universe uses current constituents, not a point-in-time survivorship-free panel. Absolute Sharpes are optimistic demo numbers, not a deployment claim. Prefer `synth` offline. DSR/PBO only correct multiple testing within the supplied panel. Not trading advice.

Reproduce

```
pip install -e ".[dev]"
python -m overfit_aware_signals synth
```

Repo

Endokelp_Projects / PythonProjects / overfit-aware-signals

Primary sources: Bailey and Lopez de Prado (2014), Deflated Sharpe Ratio; Bailey et al., Probability of Backtest Overfitting and CSCV; Lopez de Prado, Advances in Financial Machine Learning, ch. 7. Positioned as claim-filtering tooling, the same class of checks used in 2026 LLM strategy-to-code benchmarks, not an AFML tutorial clone.